Lead-Lag Relationships Between Regional Oil Prices, US Retail Oil Prices, and Crude Oil Prices

Ellie Huang Nikita Jayaprakash Annie Wang Aishani Pradhan
Steve Wong

Executive summary

...

1 Introduction

This analysis explores gasoline price dynamics by analyzing the relationship of American gasoline prices with crude oil prices and comparing regional gasoline prices within the United States. The main research questions are:

- How do Brent crude oil price changes pass through to U.S. retail gasoline prices, and over what time range does this occur?
- How do regional gasoline prices co-move over time? Are there lead–lag relationships between regions?

Based on the findings, changes in the price of Brent crude oil affect the price of gasoline in the US at a one-week lag. Regional gasoline prices are correlated and move in conjunction with each other, especially in the Midwest, Gulf Coast and East Coast. Gasoline prices in the West Coast and the Rockies regions have less correlation with the other regions, especially during larger shocks,implying more local impacts to the price. The West Coast and Rockies regions may have higher transportation-related costs and policy-driven costs, such as tax. Forecasts predict a decrease in gasoline and Brent crude oil prices in the short term with the best prediction being the mean in the long term.

1.1 Data

ADD CITATIONS

The main data sources are the weekly U.S. gasoline prices from the U.S. Energy Information Administration and Brent crude oil prices from Yahoo Finance. The weekly U.S. gasoline prices include gasoline prices for overall country and regional prices. The time period for the data is from 1994 to 2025 at a weekly frequency. This is a total of 1722 observations for each time series. Brent crude is the main input cost for gasoline, so changes in oil prices often pass through to pump prices with a lag. Regional gasoline data also captures differences in refining capacity, transportation, and local supply constraints. The regions are East Coast, Midwest, Gulf Coast, Rocky Mountains, and West Coast. This analysis helps explain how crude oil shocks affect consumer fuel prices and inflation. Forecasting this relationship can improve pricing expectations, support policy and business planning, and help anticipate the impact of supply disruptions or geopolitical events.

2 Methods

All time series (i.e. Brent crude oil prices, US gasoline prices, and the five regional gas prices) are log-transformed and differences in order to achieve stationarity. Then, the CCF (cross-correlation function) plots

are examined in order toidentify the lags with the strongest relationships and potential lead–lag structure. After fitting an initial ARMA (autoregressive moving average) model to the gasoline prices, all time series were modeled using VAR (vector autoregressive) models in order to understand and forecast the dynamics between US gasoline prices, regional gasoline prices and crude oil prices.

This analysis fit two VAR models: one using US gasoline prices and Brent crude oil prices and one using the 5 regional oil prices. These two models are used to answer the two research questions above, respectively. The appropriate lag-order for the VAR models was chosen based on several selection criteria, including AIC (Akaike Information Criterion), HQ (Hannan-Quinn Criterion), SC (Schwarz Criterion), and FPE (Final Prediction Error). In both analyses, the simplest order was chosen due to parsimony and model diagnostics. The cross-lag coefficients of the VAR models were interpreted in order to identify which series lead and which follow. Diagnostics indicated that the VAR model was appropriate for these time series. These models also are used to forecast each time series over a four-week period.

3 Results

3.1 Exploratory Data Analysis

As shown in Figure 1, regional gasoline prices, Brent crude oil prices and U.S. gasoline prices have similar shocks and directional trend, but there are still discrepancies between different time series. In particular, there are lags between the gasoline and crude oil prices, and the regional prices have varying magnitude for shocks, implying local dynamics influencing the regions individually. Some of the shocks can be explained by the historical events at the time. For example, California implemented higher gasoline taxes per gallon throughout the 2010s resulting in the dip around 2016 in the regional plot. In addition, the sharp dip in the US gasoline prices at 2008 are due to the financial crisis and the rise around 2022 are due to sanctions on Russia due to Ukraine war.

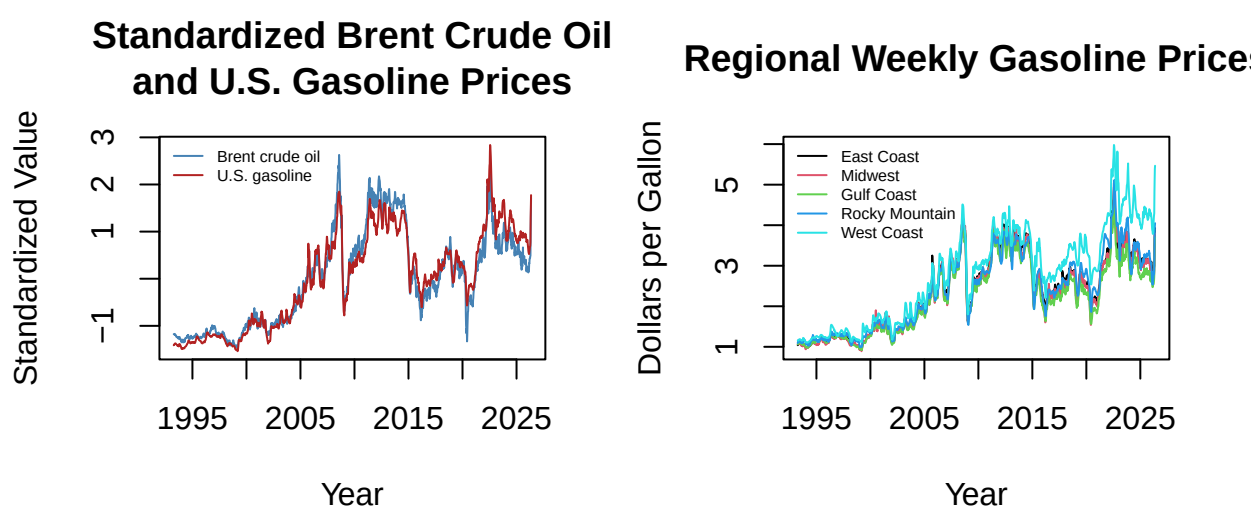

Figure 1: Regional gasoline prices, Brent crude oil prices and U.S. gasoline prices follow a shared general trend with similar rises and drops, there are still discrepancies between different time series.

3.2 Gasoline Prices AR(2) Model

As plotted in Figure 2, the fitted AR(2) model for US gasoline prices is: $gas_t = 0.5395 \times gas_{t-1} + 0.0696 \times gas_{t-2} + w_t$, where w_t is Gaussian white noise. Most of the predictable signal comes from the previous week

rather than two weeks back. The fitted mean is 0.0006, which is close to zero after differencing and logging. The residual lag 1 autocorrelation is -0.0049, so there is little obvious linear dependence left in the residuals.

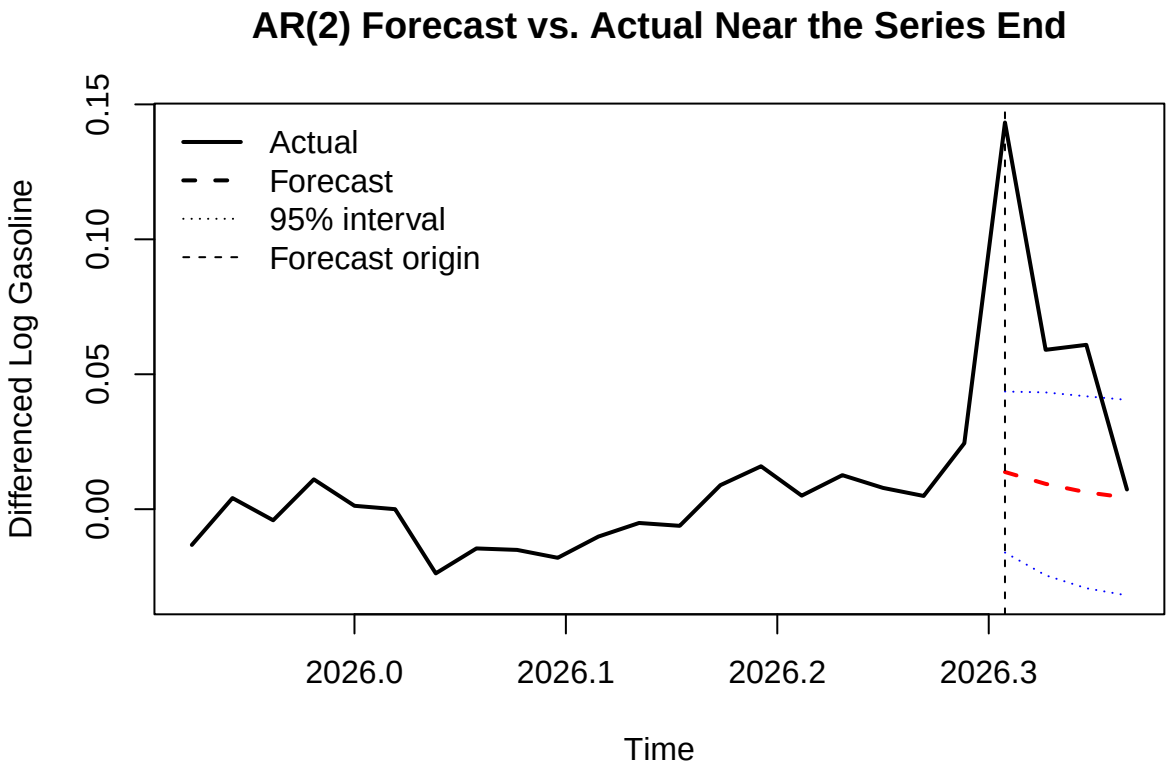

Figure 2: The AR(2) fit suggests that gasoline price changes are driven mostly by very recent history, but ultimately regress to the mean.

3.3 Gasoline-Oil Prices VAR(2) Model

Based on Figure 3, the cross-correlation function peaks at lag 0, which means gasoline and Brent often move strongly within the same week. There are also positive correlations at negative lags, especially around -1 to -3, which is consistent with Brent changes showing up in gasoline prices over the following weeks.

AIC(n)	HQ(n)	SC(n)	FPE(n)
5	3	2	5

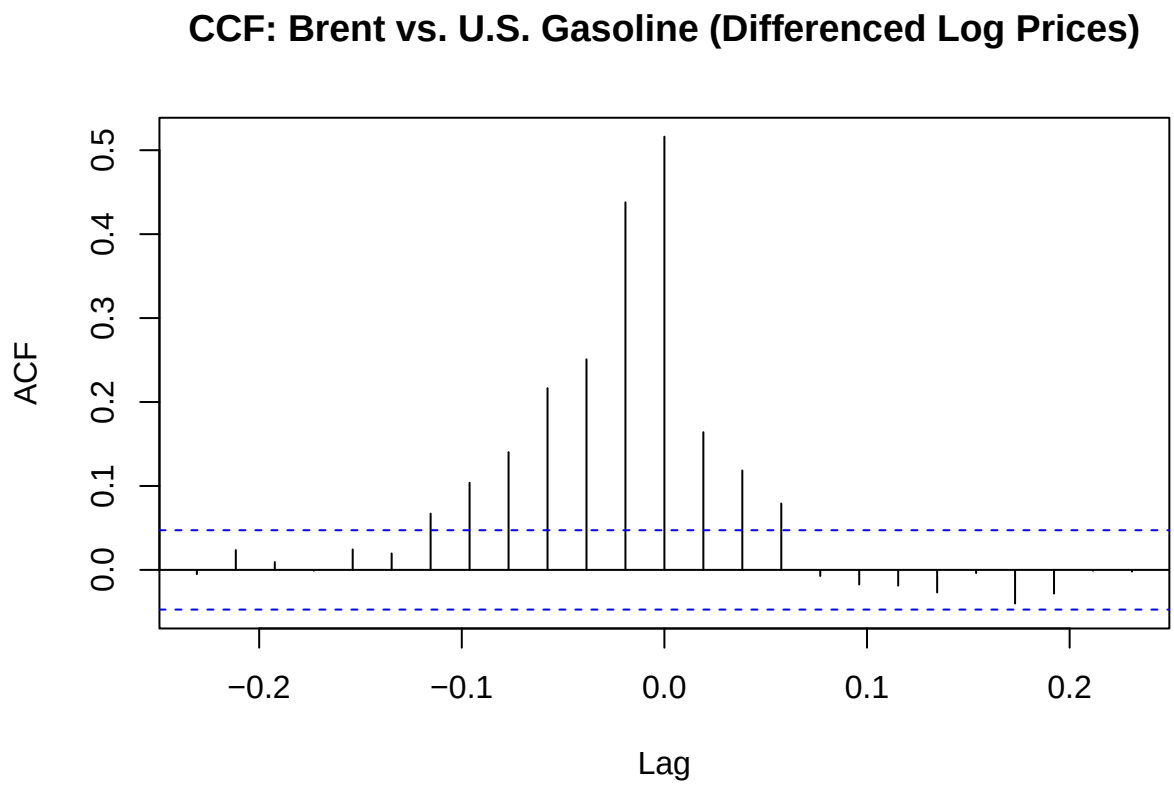

Figure 3: Label goes here

3.4 Regional Gasoline Prices VAR(1) Model

4 Discussion

4.1 AR(2) Model

The forecast plot in Figure 2 show the main limitation of this univariate model. The 4-week forecasts quickly revert toward zero and miss the sharp movement at the end of the sample. Even though the observed values remain inside the forecast interval, the point forecasts are too smooth to track the sudden change. This suggests that a model based only on past gasoline prices is not enough when market conditions shift quickly.

4.2 Diagnostics - VAR(2) Model

The largest residuals for US gasoline prices in Figure 4 are due to external shocks that are outside the scope of this model. The first peak in October 2005 is due to Hurricane Katrina, which damaged oil refineries in the gulf coast, leading to spikes in gasoline prices. The shock at November 2008 is due to the financial crisis, which decreased gasoline demand and lead to lower prices. The residual at October 2017 corresponds to OPEC announcing supply cut for oil, leading to higher prices. In April 2022, sanctions on Russian oil lead to

higher prices. Finally, during April 2026, oil prices have increased since the Straight of Hormuz is blocked. The residuals for Brent oil prices and regional gas prices have similarly timed residuals, corresponding to the same events.

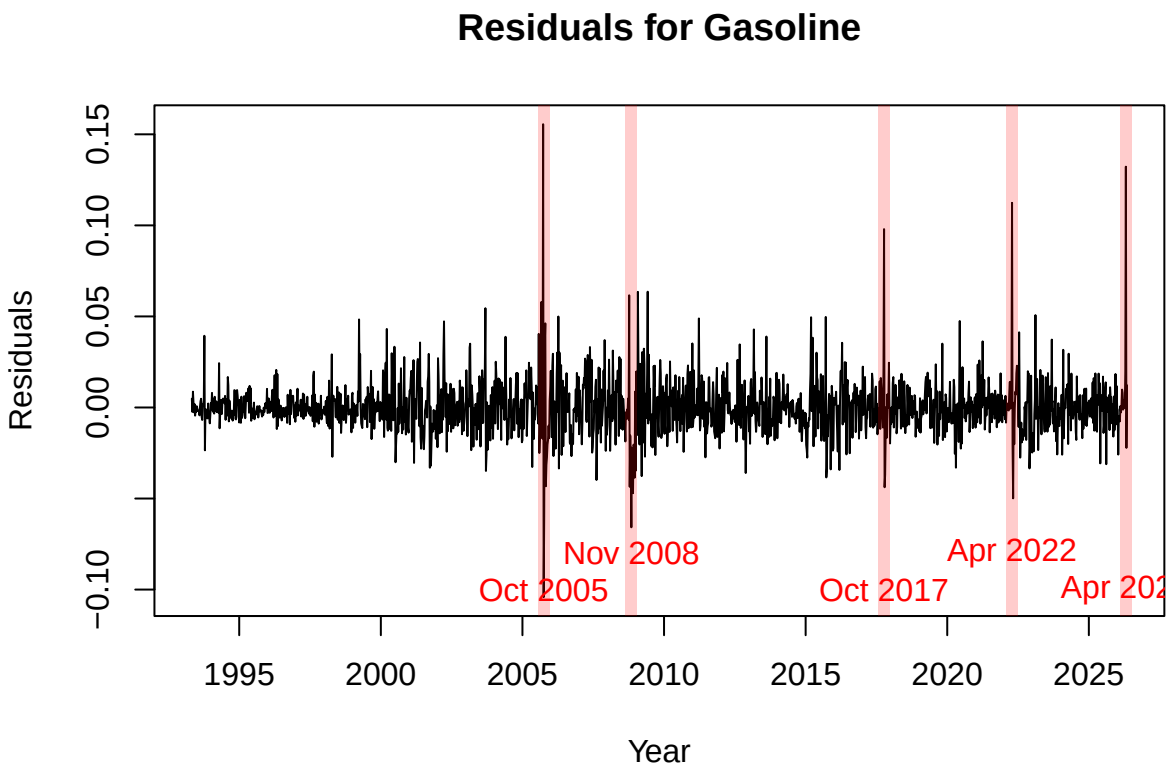

Figure 4: The largest residuals correspond to historical events that cannot be reliably modeled.

4.3 Diagnostics - VAR(1) Model

...

Appendix

Notice how the appendix below gathers all the code blocks above and nicely pastes them together.

```
###########################
# STYLE EDITS: IGNORE THIS
###########################

# normally you'll want to include this with the libraries at the beginning of your document
knitr::opts_chunk$set(message = FALSE) # include this if you don't want markdown to knit messages
knitr::opts_chunk$set(warning = FALSE) # include this if you don't want markdown to knit warnings
knitr::opts_chunk$set(echo = FALSE) # set echo = FALSE to hide code from output

###########################
# DATA CLEANING
###########################
library(here)
library(readr)
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(GGally)
library(vars)
library(forecast)
library(lubridate)
library(tibble)
# File paths
gas_file_path <- here("data", "PET_PRI_GND_A_EPM0_PTE_DPGAL_W.xls")
brent_file_path <- here("data", "brent_crude_weekly_1993_present.csv")
# Load weekly U.S. gasoline prices
us_gas_df <- read_excel(gas_file_path, sheet = "Data 1", skip = 2) |>
  rename(
    Date = 1,
    US_All_Grades = 2
  ) |>
  mutate(
    Date = as.Date(Date),
    US_All_Grades = as.numeric(US_All_Grades)
  ) |>
  arrange(Date)

# Load weekly regional gasoline prices
region_gas_df <- read_excel(gas_file_path, sheet = "Data 2", skip = 2) |>
  rename(
    Date = 1,
    East_Coast = 2,
    New_England = 3,
    Central_Atlantic = 4,
    Lower_Atlantic = 5,
    Midwest = 6,
    Gulf_Coast = 7,
    Rocky_Mountain = 8,
    West_Coast = 9,
    West_Coast_ex_CA = 10
```

```r
  ) |>
  mutate(
    Date = as.Date(Date),
    across(-Date, as.numeric)
  ) |>
  arrange(Date)

# Keep the five regions used in the regional VAR
gas_df <- us_gas_df |>
  left_join(
    region_gas_df |>
      dplyr::select(Date, East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast),
    by = "Date"
  ) |>
  drop_na()
# Load weekly Brent crude oil prices
brent_df <- read_csv(brent_file_path, show_col_types = FALSE) |>
  rename(Brent = Brent_Price_USD) |>
  mutate(
    Date = as.Date(Date),
    Brent = as.numeric(Brent)
  ) |>
  arrange(Date)
# Helper to create weekly ts objects
make_weekly_ts <- function(x, start_date) {
  ts(
    x,
    start = c(
      as.numeric(format(start_date, "%Y")),
      as.numeric(format(start_date, "%U")) + 1
    ),
    frequency = 52
  )
}

gas_start_date <- min(gas_df$Date)

# Univariate gasoline series
us_gas_ts <- make_weekly_ts(gas_df$US_All_Grades, gas_start_date)
east_ts <- make_weekly_ts(gas_df$East_Coast, gas_start_date)
midwest_ts <- make_weekly_ts(gas_df$Midwest, gas_start_date)
gulf_ts <- make_weekly_ts(gas_df$Gulf_Coast, gas_start_date)
rocky_ts <- make_weekly_ts(gas_df$Rocky_Mountain, gas_start_date)
west_ts <- make_weekly_ts(gas_df$West_Coast, gas_start_date)

# Multivariate gasoline series for regional VAR analysis
region_var_ts <- gas_df |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  as.matrix() |>
  ts(
    start = c(
      as.numeric(format(gas_start_date, "%Y")),
      as.numeric(format(gas_start_date, "%U")) + 1
```

```r
    ),
    frequency = 52
  )

colnames(region_var_ts) <- c(
  "East_Coast", "Midwest", "Gulf_Coast", "Rocky_Mountain", "West_Coast"
)
# Align gasoline and Brent crude on common weekly dates
gas_brent_df <- gas_df |>
  inner_join(brent_df, by = "Date") |>
  drop_na()
gas_brent_start_date <- min(gas_brent_df$Date)

gas_ts_brent_overlap <- make_weekly_ts(gas_brent_df$US_All_Grades, gas_brent_start_date)
brent_ts <- make_weekly_ts(gas_brent_df$Brent, gas_brent_start_date)

###########################
# EXPLORATORY DATA ANALYSIS
###########################
par(mfrow = c(1,2))
# Standardize gasoline and Brent for visual comparison
gas_scaled_ts <- ts(
  as.numeric(scale(as.numeric(gas_ts_brent_overlap))),
  start = start(gas_ts_brent_overlap),
  frequency = frequency(gas_ts_brent_overlap)
)

brent_scaled_ts <- ts(
  as.numeric(scale(as.numeric(brent_ts))),
  start = start(brent_ts),
  frequency = frequency(brent_ts)
)

ts.plot(
  brent_scaled_ts, gas_scaled_ts,
  col = c("steelblue", "firebrick"), lty = 1,
  main = "Standardized Brent Crude Oil\nand U.S. Gasoline Prices",
  xlab = "Year",
  ylab = "Standardized Value"
)
legend(
  "topleft",
  legend = c("Brent crude oil", "U.S. gasoline"),
  col = c("steelblue", "firebrick"),
  lty = 1,
  bty = "n",
  cex = 0.5
)
# Regional gasoline prices
ts.plot(
  east_ts, midwest_ts, gulf_ts, rocky_ts, west_ts,
  col = 1:5, lty = 1,
  main = "Regional Weekly Gasoline Prices",
```

```r
  xlab = "Year",
  ylab = "Dollars per Gallon"
)
legend(
  "topleft",
  legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
  col = 1:5, lty = 1, bty = "n",
  cex = 0.5
)
############################
# GASOLINE AR MODEL
############################
# Difference log gasoline prices to work with an approximately stationary series
d_us_gas <- diff(log(us_gas_ts))

# Hold out the last 4 weeks for a simple forecast check
forecast_horizon <- 4
n_gas <- length(d_us_gas)

train_d_us_gas <- window(d_us_gas, end = time(d_us_gas)[n_gas - forecast_horizon])
test_d_us_gas <- window(d_us_gas, start = time(d_us_gas)[n_gas - forecast_horizon + 1])

# Fit AR(2) model
fit_ar2_gas <- arima(train_d_us_gas, order = c(2, 0, 0))

# Forecast the held-out period
fc_ar2_gas <- forecast(fit_ar2_gas, h = forecast_horizon, level = 95)

fc_mean_ts <- ts(
  as.numeric(fc_ar2_gas$mean),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

fc_lower_ts <- ts(
  as.numeric(fc_ar2_gas$lower[, 1]),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

fc_upper_ts <- ts(
  as.numeric(fc_ar2_gas$upper[, 1]),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

# Show recent history leading into the forecast origin
weeks_of_history <- 20
tail_start_time <- time(d_us_gas)[n_gas - forecast_horizon - weeks_of_history + 1]
actual_tail_ts <- window(d_us_gas, start = tail_start_time)

plot(
  actual_tail_ts,
```

```r
  type = "l",
  lwd = 2,
  main = "AR(2) Forecast vs. Actual Near the Series End",
  xlab = "Time",
  ylab = "Differenced Log Gasoline",
  ylim = range(c(actual_tail_ts, fc_mean_ts, fc_lower_ts, fc_upper_ts))
)
lines(fc_mean_ts, lwd = 2, lty = 2, col = "red")
lines(fc_lower_ts, lty = 3, col = "blue")
lines(fc_upper_ts, lty = 3, col = "blue")
abline(v = time(test_d_us_gas)[1], lty = 2)
legend(
  "topleft",
  legend = c("Actual", "Forecast", "95% interval", "Forecast origin"),
  lty = c(1, 2, 3, 2),
  lwd = c(2, 2, 1, 1),
  bty = "n"
)
# First differences of log prices
d_us_gas_overlap <- diff(log(gas_ts_brent_overlap))
d_brent <- diff(log(brent_ts))

# Calendar dates corresponding to the differenced series
gas_brent_diff_dates <- gas_brent_df$Date[-1]
# Negative lag means Brent leads gasoline
ccf(
  d_brent, d_us_gas_overlap,
  lag.max = 12,
  main = "CCF: Brent vs. U.S. Gasoline (Differenced Log Prices)"
)
# Bivariate stationary series for VAR analysis
oil_var_ts <- cbind(
  Gasoline = d_us_gas_overlap,
  Brent = d_brent
)
var_analysis <- VARselect(oil_var_ts, lag.max = 8, type = "none")
# Candidate lag lengths
gt::gt(as.data.frame(t(var_analysis$selection),
                     check.names = F))
fit_var_oil <- VAR(oil_var_ts, p = 2, type = "none")
resid_oil  <- resid(fit_var_oil)
plot_residuals_with_events <- function(res_vec, dates_vec, series_name,
                                       threshold_upper, threshold_lower,
                                       label_y_positions) {

  # Build dataframe once
  df <- data.frame(time = dates_vec,
                   value = res_vec)

  # Filter event dates
  event_dates <- df |>
    filter(value > threshold_upper | value < threshold_lower) |>
    group_by(floor(time)) |>
```

```r
    summarize(dates = mean(time), .groups = "drop") |>
    mutate(dates = as.Date(date_decimal(dates))) |>
    pull(dates)

  # Convert to decimal time
  event_times <- as.numeric(format(event_dates, "%Y")) +
    (as.numeric(format(event_dates, "%j")) - 1) / 365

  # Plot
  ts.plot(ts(res_vec,
             start = dates_vec[1],
             frequency = 52),
          lty = 1,
          main = paste("Residuals for", series_name),
          xlab = "Year", ylab = "Residuals")

  usr <- par("usr")

  # Draw rectangles (vectorized)
  rect(xleft  = event_times - 0.2,
       xright = event_times + 0.2,
       ybottom = usr[3],
       ytop    = usr[4],
       col = rgb(1, 0, 0, alpha = 0.2),
       border = NA)

  # Labels
  y_positions <- rep(label_y_positions, length.out = length(event_times))

  text(x = event_times,
       y = y_positions,
       labels = format(event_dates, "%b %Y"),
       col = "red",
       cex = 1)
}

# Gasoline
plot_residuals_with_events(
  res_vec = resid_oil[,1],
  dates_vec = time(oil_var_ts)[-c(1,2)],
  series_name = "Gasoline",
  threshold_upper = 0.08,
  threshold_lower = -0.05,
  label_y_positions = c(-0.1, -0.08)
)
```